

Solution Requirements

Date	15 JULY 2026
Team ID	SWTID-2026-2749
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Admin login to access the dashboard securely.	High
2	Authorization levels	Admin manages the application while farmers use crop prediction.	Medium
3	External interfaces	Flask web application interacts with the trained Random Forest model.	High

4	Transactions processing	Accepts NPK values, temperature, humidity, pH, and rainfall to generate crop recommendations.	High
5	Reporting	Displays recommended crop with season, soil type, temperature range, and water requirement.	High
6	Business rules, etc.	Prediction is generated only after all required agricultural parameters are entered.	High
7	Compliance to laws or regulations	Protects user input and follows secure programming practices.	Medium
8	<i>Other</i>	Allows easy future integration of fertilizer and disease prediction modules.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Prediction should be generated quickly.	Response within 2 seconds
2	Scalability	System should support additional crops and larger datasets.	Supports future dataset expansion
3	Security & Data Privacy	Validate user inputs before prediction.	No invalid inputs accepted
4	Reliability & Availability	Prediction should be available whenever the application is running.	99% availability during operation
5	Usability & Accessibility	Simple interface suitable for farmers.	Easy navigation and responsive design
6	<i>Other</i>	Maintainable modular code.	Easy to update models and features